Astrometrica on Apple Silicon macOS for Asteroid Search: Installation, Validation, and Workflow

Aditya Rajiv Ratnam
National Academy for Learning, Bengaluru
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adityaratnam09 AT gmail DOT com

Abstract

This report documents a verified procedure for running Astrometrica, the astrometric reduction software used in International Astronomical Search Collaboration (IASC) campaigns, on Apple Silicon macOS systems. Because Astrometrica is a legacy 32-bit x86 Windows application that has not been updated for modern operating systems, a Windows compatibility layer is required to execute it on current Apple hardware. Several compatibility approaches were investigated experimentally, after which Sikarugir, an open-source Wine wrapper used in combination with Rosetta 2, was selected as the most reliable option. The principal contribution of this report is the successful execution of the complete asteroid search workflow on Apple Silicon hardware. Installation of the required software, image loading, astrometric data reduction, moving object detection, known object cross referencing, manual candidate search, and reporting to the Minor Planet Center are each presented in sequence and independently verified by manual execution of every step, establishing that the full IASC pipeline runs reliably in a translated environment without access to native Windows. The procedure successfully detected a moving object in an official IASC practice dataset, demonstrating end-to-end operation of the workflow.

Keywords: Astrometrica, Apple Silicon, macOS, Wine compatibility, Rosetta 2, Sikarugir, IASC, Pan-STARRS, asteroid search, astrometric reduction, Minor Planet Center, citizen science, FITS

1. Introduction

1.1 Asteroids and Minor Planets

The solar system formed approximately 4.6 billion years ago from the gravitational collapse of dust and gas surrounding the young Sun. Most of this material accreted into planets; the remainder persisted as smaller bodies known collectively as minor planets, comprising asteroids, comets, and dwarf planets. Asteroids are predominantly rocky bodies ranging from a few meters to several hundred kilometers in diameter. Vesta, at approximately 530 km, is the largest asteroid not classified as a dwarf planet; Ceres, at approximately 940 km, is sufficiently large and spherical to meet the International Astronomical Union criteria for dwarf planet status (International Astronomical Union, 2006).

The majority of asteroids occupy the Main Belt, located between the orbits of Mars and Jupiter and stabilized largely by Jupiter's gravitational influence. A smaller population, the Trojan asteroids, share Jupiter's orbit at gravitationally stable Lagrange points leading and trailing the planet. A further population exists beyond the orbit of Neptune.

Asteroids are of scientific interest for two principal reasons. Their composition provides evidence of the conditions under which the solar system formed because they formed early in the history of the solar system and have undergone comparatively little subsequent alteration. In addition, a subset of asteroids, termed Near Earth Objects (NEOs), cross Earth's orbit and are therefore directly relevant to impact risk assessment. Each orbit measured and catalogued contributes to both objectives, which constitutes the underlying scientific rationale for citizen science search campaigns of the kind described in this report.

1.2 The IASC Campaign and Pan-STARRS

The International Astronomical Search Collaboration (IASC) partners with the National Aeronautics and Space Administration Jet Propulsion Laboratory (NASA JPL) and the Minor Planet Center (MPC) to provide citizen scientists with access to unprocessed survey images rather than simulated data (International Astronomical Search Collaboration [IASC], 2026). These images originate from Pan-STARRS, the Panoramic Survey Telescope and Rapid Response System, located on Haleakalā in Maui, Hawaii, and among the more prolific NEO discovery telescopes currently in operation (Kaiser et al., 2010). Two units, designated PS1 and PS2, generate the campaign image sets, which are labeled accordingly. Matching the Astrometrica configuration file (PS1.cfg or PS2.cfg) to the corresponding prefix of the image set is required for correct processing, as discussed in Section 2.5.

Each image set consists of four exposures of the same region of sky, typically captured at intervals of 15 to 20 minutes. Campaign participants are generally organized into teams, with individual members processing separate image sets in parallel, since manual review of a set requires substantial time and thorough sky coverage is prioritized over the speed of any individual reviewer.

1.3 Detection and Reporting Pipeline

Following submission of a report, a candidate proceeds through a defined review pipeline maintained by the Minor Planet Center and NASA JPL. This pipeline is summarized below, as it establishes the context in which the results of this study should be interpreted.

- **Preliminary detection (approximately one week).** The Minor Planet Center performs an initial assessment of the submission to determine whether the reported observation is consistent with a genuine moving object rather than noise or an existing catalog entry.

- **Provisional detection (approximately eight months to one year).** Following a successful preliminary assessment, the Minor Planet Center tracks the same region of sky over an extended period to establish a preliminary orbit, confirming that the object follows a consistent and predictable path.
- **Numbered discovery (three to five years).** Once a preliminary orbit is established, NASA JPL continues to track the object to confirm orbital stability, after which a permanent designation number and a complete set of orbital elements, including eccentricity, inclination, and semi-major axis, are assigned and published in the JPL Small-Body Database.

Only a small proportion of preliminary detections proceed through the full pipeline to numbered discovery, reflecting the multi-year duration of orbit confirmation. The candidate identified in the present study is obtained from a practice image set rather than a live campaign submission. It is not submitted for Minor Planet Center review. A complete technical analysis, structured in the manner of a formal detection report, is planned upon application of this procedure to live campaign data.

2. Methodology

The procedure below is presented in two parts: Sections 2.1 to 2.4 cover Apple Silicon installation and configuration, while Sections 2.5 onward describe standard Astrometrica operation applicable to any platform, with macOS-specific considerations noted where relevant.

2.1 Selection of a Windows Compatibility Layer

Astrometrica is a legacy 32-bit x86 Windows application (Astrometrica Official Documentation, n.d.) and requires a translation layer to execute on Apple Silicon hardware. Four candidate approaches are evaluated prior to selection, summarized in Table 1.

Approach	Description	Assessment
Sikarugir	A free, open-source Wine wrapper (Sikarugir Project, 2026), the community successor to Wineskin, used in combination with Rosetta 2 to translate Intel instructions.	Selected for this procedure. Lightweight, provides direct Finder access to program files, and does not require management of a separate virtual machine.
Whisky	A newer Wine wrapper developed primarily for modern 64-bit DirectX games.	Not selected. No longer maintained by its developer. Failed to install dependencies such as WhiskyWine correctly.

Porting Kit	A Wine bundler with automated installation wizards for specific supported titles.	Not selected. Astrometrica is not among its supported titles, and crashes on startup.
Full Windows virtual machine (UTM, VirtualBox, Parallels)	Runs a complete Windows installation inside a virtual machine.	Functional but resource-intensive: requires 20 to 40 GB of disk space for a full operating system to run a single lightweight program, with a noticeable additional load on battery life and memory.

Table 1. Comparison of Windows compatibility approaches evaluated for Apple Silicon macOS.

2.2 Installation of Homebrew, Sikarugir, and Rosetta 2

Three software components are required prior to installation of Astrometrica: Homebrew, used to retrieve the remaining components; Sikarugir, the Wine wrapper used to execute Astrometrica; and Rosetta 2, the Apple translation layer on which Sikarugir depends. The following commands are executed in Terminal, in sequence.

Tested configuration: macOS 15.7.4, Apple M1 chip, Sikarugir version 1.0.1, Astrometrica version 4.10.6.453.

2.2.1 Homebrew Installation

Homebrew serves as the package manager used to retrieve the remaining software components.

```
/bin/bash -c "$(curl -fsSL
https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
echo >> $HOME/.zprofile
echo 'eval "$(/opt/homebrew/bin/brew shellenv zsh)"' >> $HOME/.zprofile
eval "$(/opt/homebrew/bin/brew shellenv zsh)"
brew upgrade
```

2.2.2 Sikarugir Installation

Homebrew casks originating from outside the default repositories must be explicitly trusted prior to installation; this is performed by the first command below.

```
brew trust Sikarugir-App/sikarugir
brew install --cask Sikarugir-App/sikarugir/sikarugir
```

Note. *Additional flags such as '--no-quarantine' and installation from unofficial taps are not recommended, as they can interfere with Gatekeeper security checks on current macOS versions and cause installation failures.*

2.2.3 Rosetta 2 Installation

Rosetta 2 is the Apple translation layer required to execute Intel compiled code on Apple Silicon hardware, and is a dependency of Sikarugir.

```
softwareupdate --install-rosetta --agree-to-license
```

2.3 Configuration of Sikarugir for Astrometrica

- Sikarugir Creator is opened from the Applications folder.
- On first launch, its template wrapper must first be downloaded. Following completion, the engine selection interface is used to select WS12WineSikarugir, the runtime that enables execution of 32-bit Windows software under Rosetta.

Figure 1. *Sikarugir Creator during download of the wrapper template, with the WS12WineSikarugir engine selected from the available runtime options.*

- The Astrometrica installer is executed through Sikarugir, producing a self-contained application bundle.
- One error encountered at this stage is a path not found error occurring when Astrometrica first attempts to save its settings. This is a common occurrence when Windows software expects local folders that do not yet exist within a newly created Wine prefix. The issue is resolved as follows:
    - The Sikarugir control panel for the Astrometrica wrapper is opened.

  - The Advanced tab is selected, followed by 'Run Wine Command Prompt (cmd)'.
  - The following command is entered to create the missing folder:

```
mkdir %localappdata%\Astrometrica
```

- The command prompt is closed, completing configuration of Sikarugir.

Note. *Where a registration key is provided by a campaign team leader, it should be entered once under Help > Registration within Astrometrica, removing further trial limitations.*

2.4 File Management

Sikarugir maps the simulated Windows drive to a standard macOS folder, permitting file access through Finder rather than through the Wine environment directly.

- The Astrometrica application icon created by Sikarugir is located.
- The icon is right-clicked and 'Show Package Contents' is selected.
- The following path is accessed: Contents > SharedSupport > prefix > drive_c > Astrometrica.

2.4.1 Image File Placement

A subfolder for campaign files (designated CampaignImages in this procedure) is created within the Astrometrica directory, and FITS files are placed within it.

Note. *FITS (Flexible Image Transport System) is the standard file format for astronomical images (Greisen & Calabretta, 2002). In addition to pixel data, FITS files carry metadata including telescope coordinates, exposure timestamps, and sensor characteristics.*

2.4.2 Orbit Catalog Updates

Astrometrica references a local copy of the Minor Planet Center orbit database to flag previously catalogued objects (Minor Planet Center, 2026). This file requires manual updating:

- The MPC Orbit database page is accessed at https://www.minorplanetcenter.net/iau/MPCORB.html.
- The file MPCOrb.dat is downloaded.
- The file is placed in drive_c/Astrometrica/Catalogs/, replacing any existing copies.

2.5 Loading of Configuration and Image Files

From this point, the procedure describes standard Astrometrica operation, independent of platform.

- Astrometrica is opened. Under Settings > Open from the toolbar (or File | Settings), the .cfg file provided by the campaign is loaded. For Pan-STARRS data, this file is typically named PS1.cfg or PS2.cfg.

  Note. *The configuration file selected must correspond to the filename prefix of the image set being processed. Selecting a mismatched configuration file (for example, PS2.cfg for a PS1 image set) is a common source of error and results in rejection of the Minor Planet Center report despite correct execution of all preceding steps.*

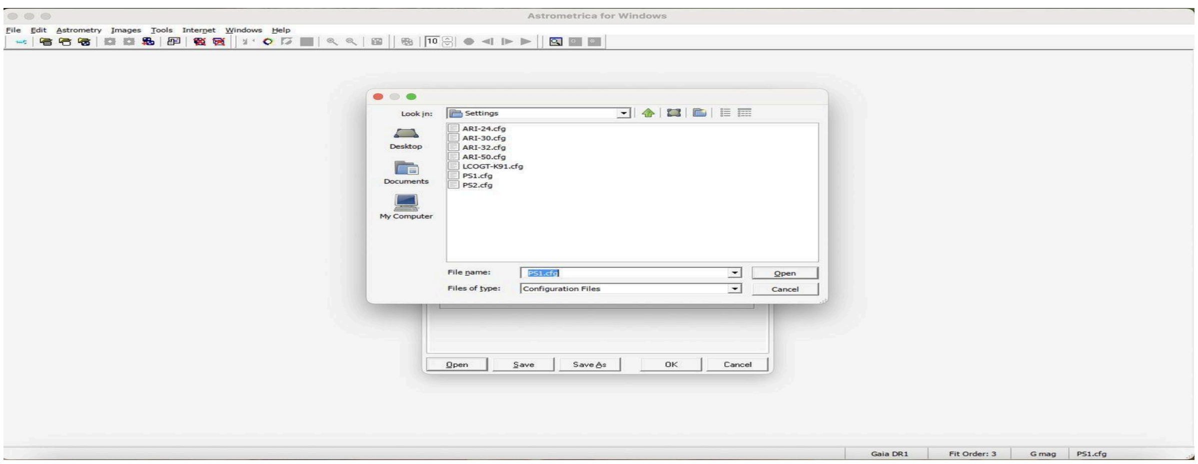

Figure 2. *The Astrometrica Settings manager, showing selection of the campaign configuration profile (PS1.cfg).*

- Image files are loaded via File > Load Images, or the shortcut Ctrl+L.

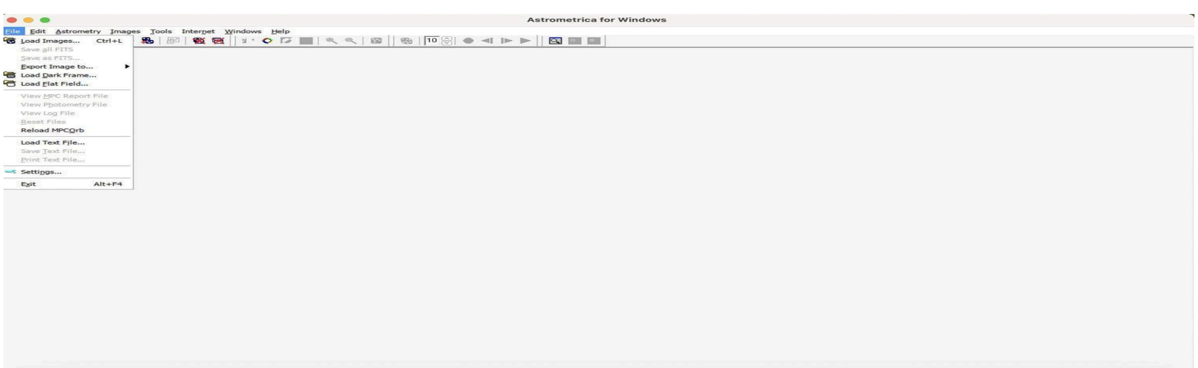

Figure 3a. *The Load Images dialog used to import the campaign FITS files.*

- Each campaign image set comprises four FITS frames of the same region of sky, captured at fixed intervals. All four frames are selected and loaded in chronological order.

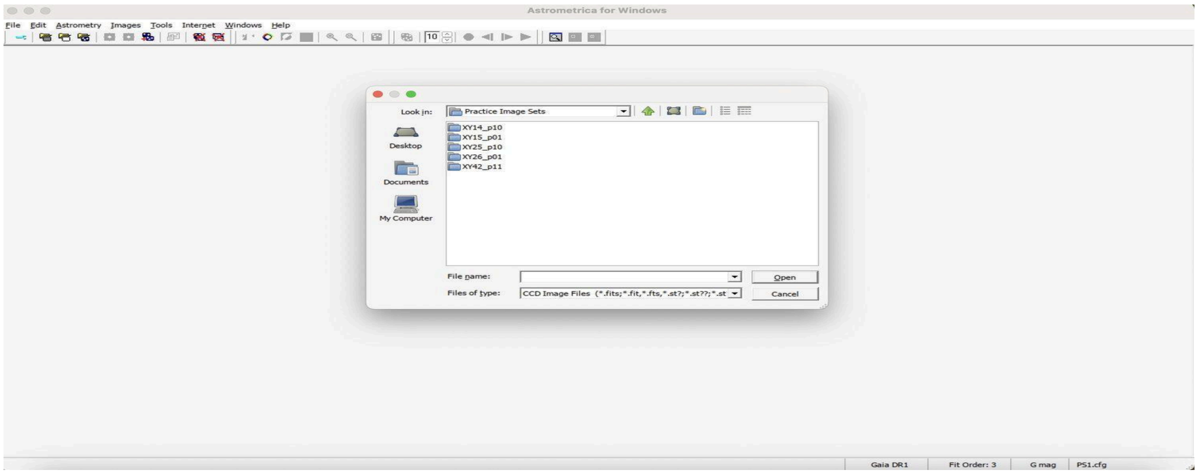

Figure 3b. *The folder containing the four-image sequence for the campaign. The Practice Image Sets directory is shown in this example.*

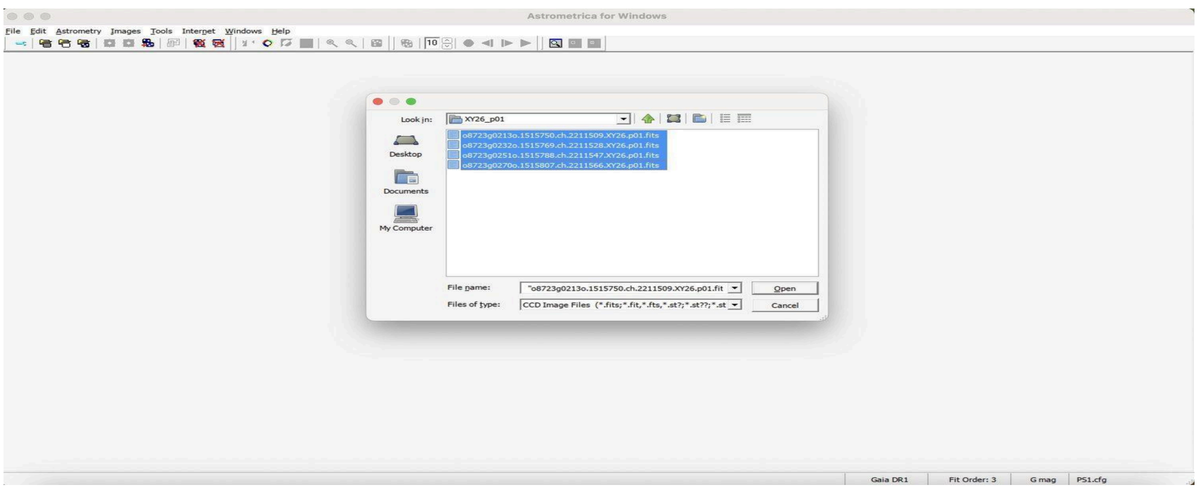

Figure 3c. *Selection of the four FITS files comprising the image set.*

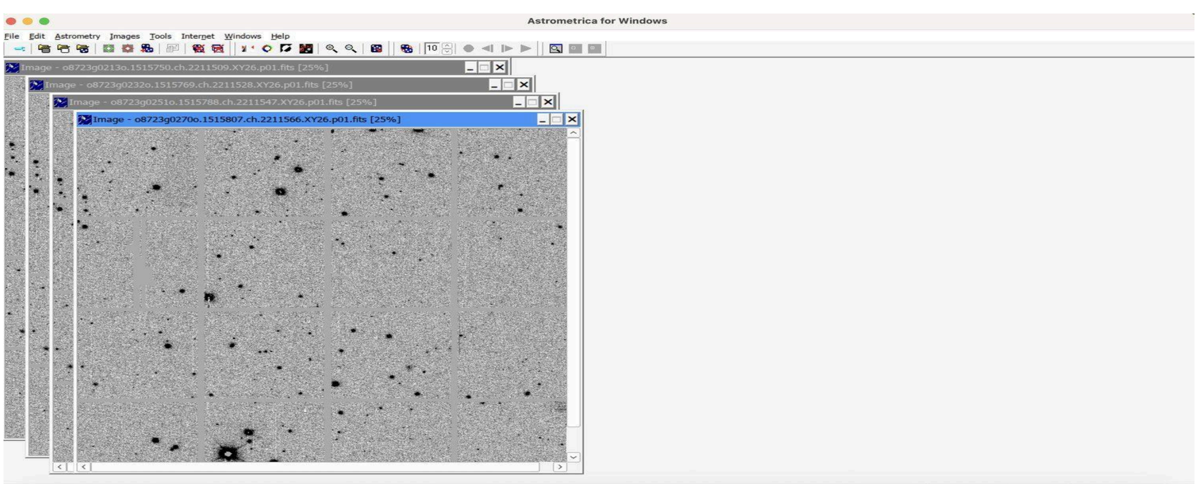

Figure 3d. *All four frames successfully loaded, each displayed in a separate image window.*

2.6 Parameter Configuration for a Translated Environment

Because Astrometrica executes through a translation layer rather than natively, default parameter values are reviewed under Program Settings (Program tab) prior to processing.

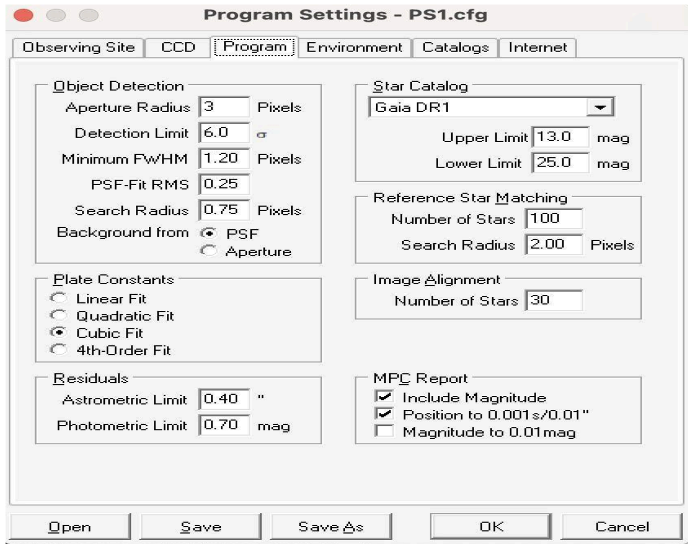

Figure 4. *The Program Settings dialog, showing the Object Detection, Plate Constants, Residuals, Star Catalog, MPC Report, and related configuration panels used for this campaign.*

- **Aperture Radius, set to 3.** This defines the pixel radius used by Astrometrica to measure stellar brightness. A narrower radius reduces the volume of background data processed, which is of particular relevance when executing through Wine.
- **Detection Limit, set to 6.0.** This is the signal to noise threshold below which a source is classified as background noise rather than a candidate star.
- **Minimum FWHM, set to 1.20.** This value excludes hot pixels and sensor artifacts, which characteristically appear sharper than genuine stellar sources.

Note. *A value of 1.20 provides a reasonable baseline for excluding single pixel hot noise on modern sensors. Pan-STARRS images have a comparatively large pixel scale, approximately 0.26 arcseconds per pixel, and the campaign supplied configuration file frequently includes its own detection limits calibrated for this scale. Where the supplied configuration specifies a different value, that value should take precedence over the baseline given here.*

- **Star Catalog.** Set according to campaign specification, typically Gaia DR1.

Note. *Object detection and data reduction may cause the status bar to appear unresponsive for five to ten minutes during a full processing run. This behavior is expected and does not indicate a fault.*

2.7 Astrometric Data Reduction

Prior to candidate identification, Astrometrica establishes the coordinates of the loaded frames by matching detected sources against a star catalog. This step is performed using the Astrometric Data Reduction function, accessible from the toolbar or via the shortcut Ctrl+A.

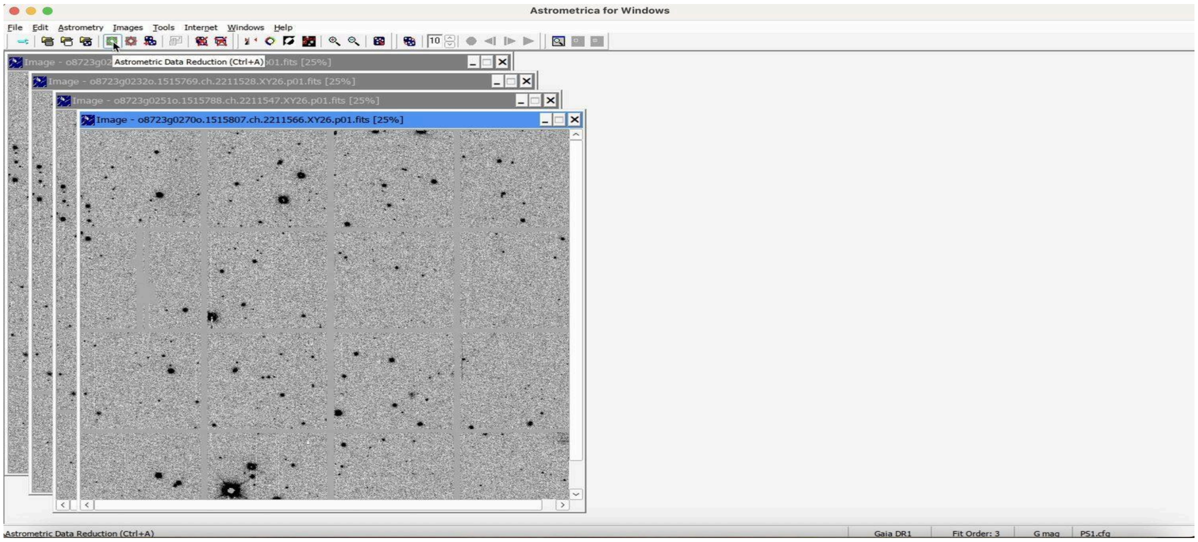

Figure 5a. *The Astrometric Data Reduction control on the Astrometrica toolbar.*

This function performs an automated plate solving procedure in which point sources within the image are matched against the Gaia catalog to determine precise coordinates (Gaia Collaboration, 2016).

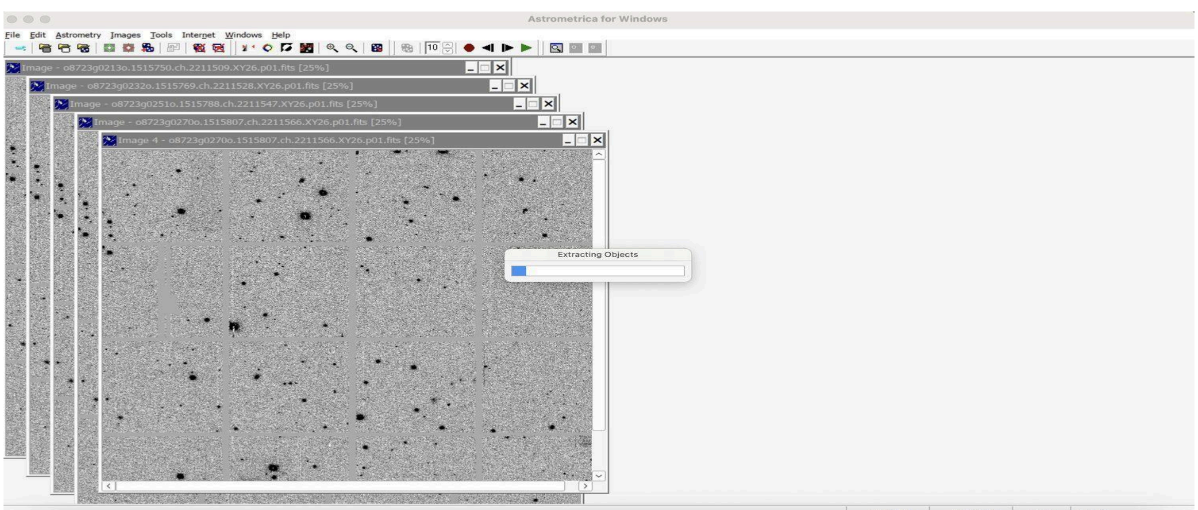

Figure 5b. *Astrometrica extracting point sources during the plate solving procedure.*

On completion, a results window is displayed reporting residuals, fit scale, and the number of stars successfully matched.

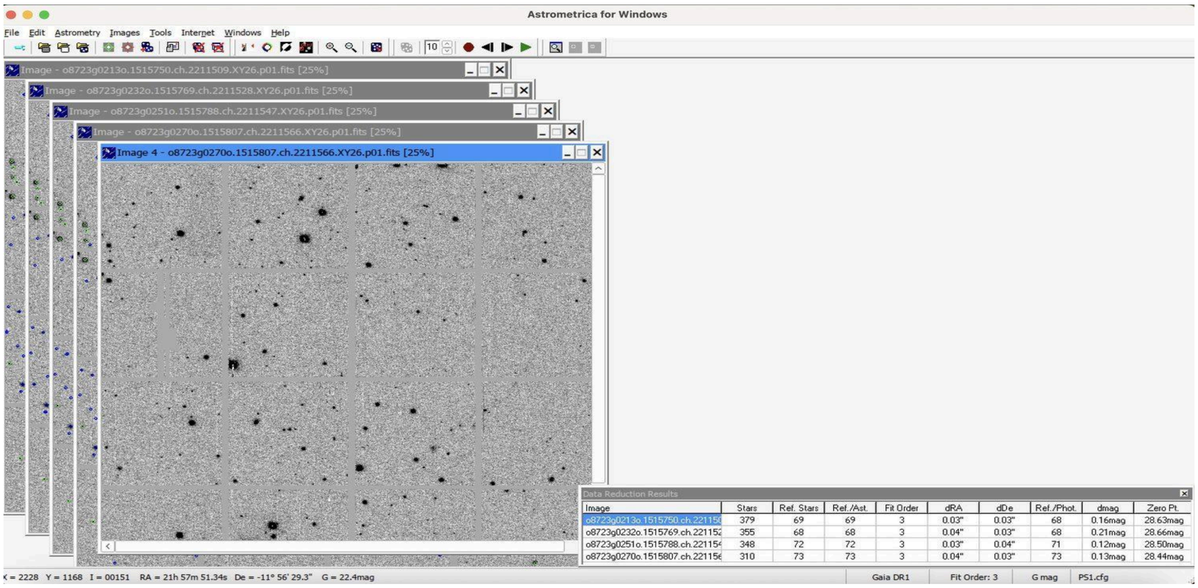

Figure 5c. *The Data Reduction Results window, showing matched star counts and fit residuals following a successful reduction.*

2.7.1 Reference Star Alignment Errors

An issue that may be encountered at this stage is a reference star alignment error, reported as 'Reference Star Match Error' or 'Less than 3 Reference Stars found'. The underlying cause is as follows: to align an image, Astrometrica selects the N brightest non-saturated stars detected in the image and attempts to match them against the N brightest catalog stars within a specified magnitude range, where N corresponds to the configured 'Number of Stars' parameter for the reference match. Where these two sets fail to overlap sufficiently, the match fails. This condition typically arises from one of the following causes.

- **Mismatch between saturation and magnitude limits.** This is the most frequent cause. Where an image saturates at approximately magnitude 12 but the catalog upper magnitude limit is set to 8, Astrometrica compares dimmer, unsaturated image stars against the brightest catalog stars, producing minimal overlap. The catalog upper magnitude limit should be set under Program Settings to approximate the actual saturation point of the images.
- **Insufficient stars within the field.** Sparse fields, such as those near the galactic poles, may not contain sufficient bright stars for a confident match. Increasing the 'Number of Stars' parameter, initially to 80 and subsequently to 160 if required, may help.
- **Crowded fields.** Densely populated star fields can similarly disrupt automated matching. In this case, a smaller and more selective star count, in the range of 30 to 50, is often more effective than a larger value.

- **Catalog substitution as a final option.** Gaia DR2 or DR3 catalogs are dense and may perform poorly in crowded fields; substitution with UCAC4, a sparser catalog, occasionally resolves a match that Gaia catalogs cannot. This is a workaround rather than a permanent correction, and the catalog should be reset for subsequent image sets.

Where none of the preceding measures resolve the error, a manual match may be performed by declining automatic detection when prompted and adjusting the star pattern overlay, using the on screen controls, until it aligns with three or four unambiguous bright stars within the frame.

2.8 Moving Object Detection

With frame coordinates established, Astrometrica performs an automated scan across the four loaded frames to identify sources whose position shifts consistently between exposures. This step precedes the Known Object Overlay described in Section 2.9, as moving sources must first be identified before their catalog status can be determined.

- The Moving Object Detection control is selected from the toolbar.

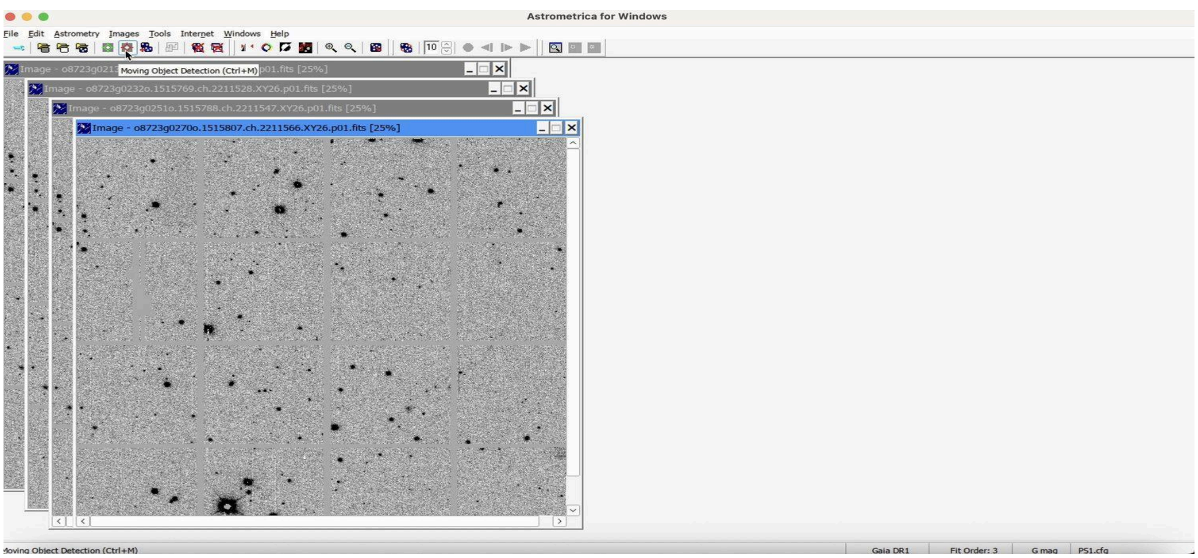

Figure 6a. *The Moving Object Detection control on the Astrometrica toolbar.*

- Selection of this control initiates an automated scan across all four frames for consistent motion. This is typically the most time-intensive step in the procedure, requiring several minutes within a translated environment.

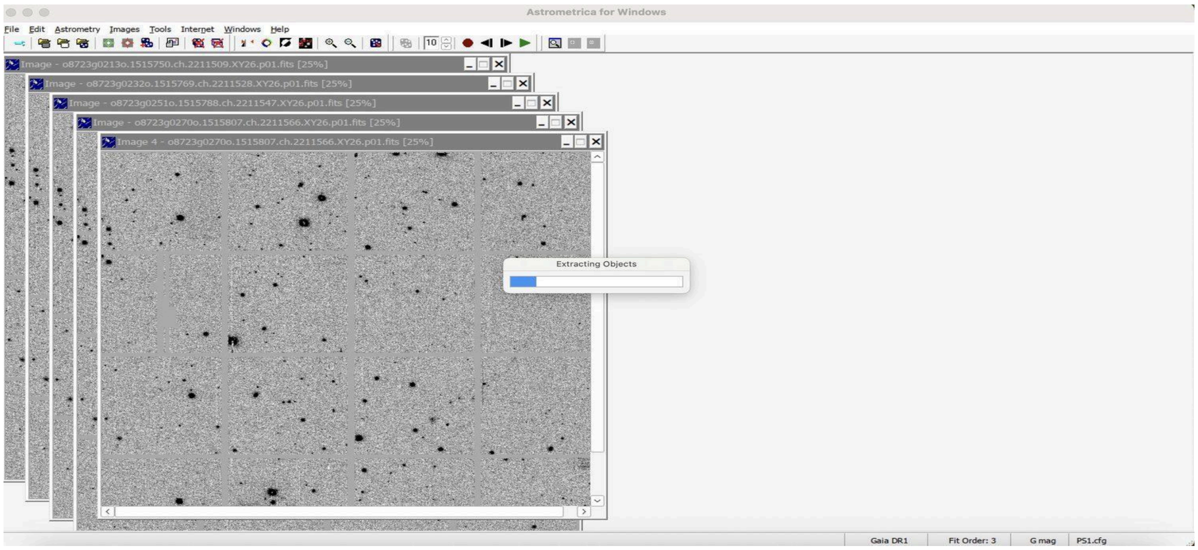

Figure 6b. *Moving Object Detection scanning across the four loaded frames for consistent motion.*

- On completion, the results list is reviewed. Sources corresponding to a previously catalogued object are flagged automatically.

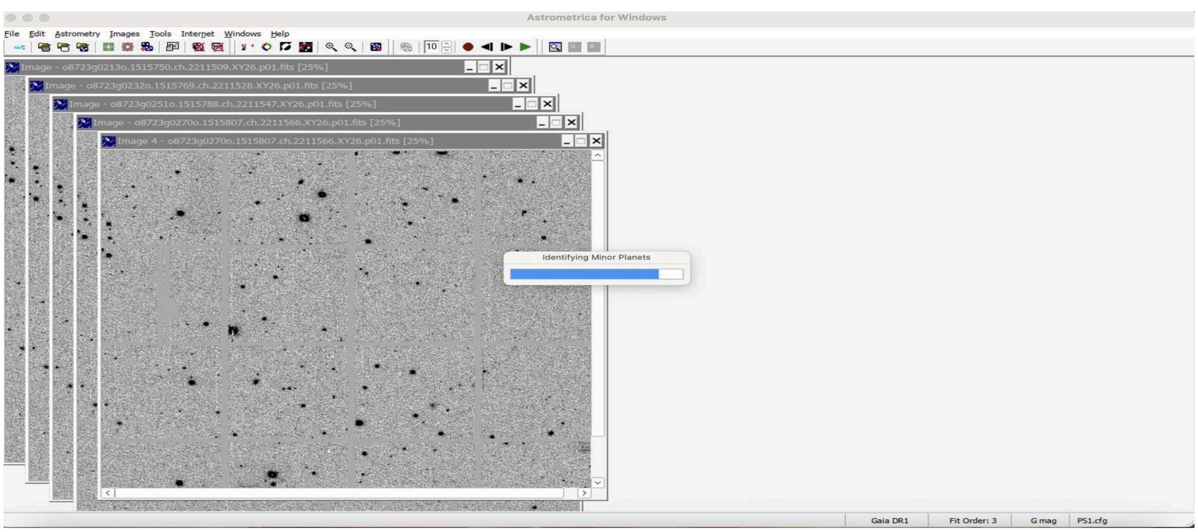

Figure 6c. *Progress dialog showing minor planet identification against the local catalog in progress, following completion of the moving object scan.*

- Where the reported trajectory corresponds to a catalogued object, the match is accepted and logged as known.

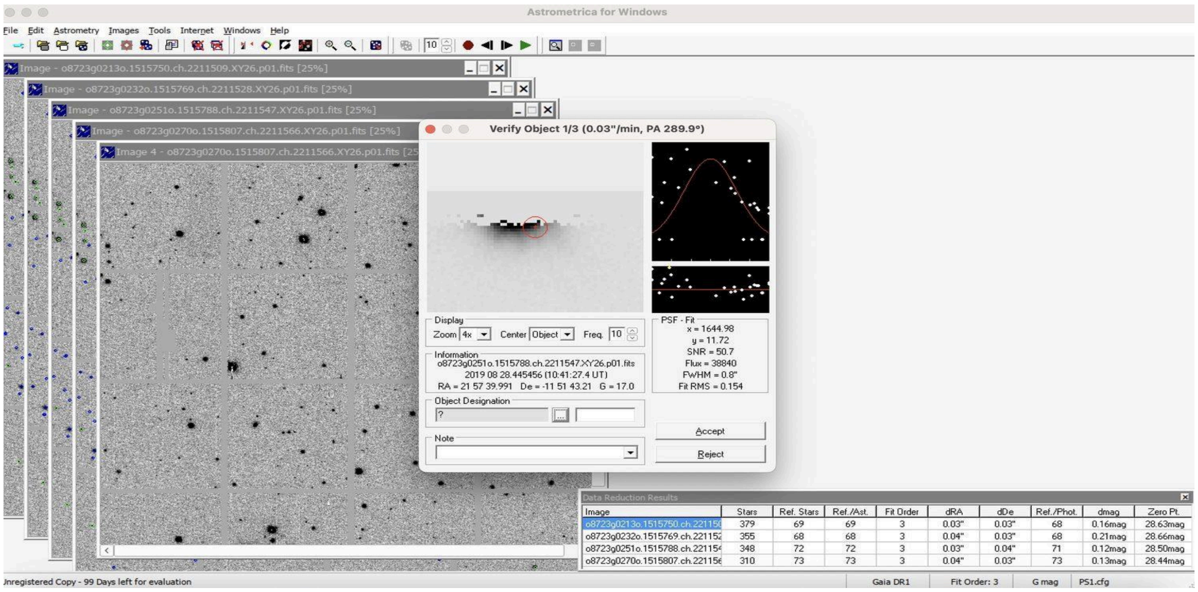

Figure 6d. *Confirmation of a candidate motion vector against the catalog prior to acceptance of the match.*

2.9 Known Object Overlay

Following identification of moving sources, this step establishes which of those sources correspond to previously catalogued objects, preventing an already documented object from being reported as a new candidate.

- The Known Object Overlay control is selected from the toolbar.

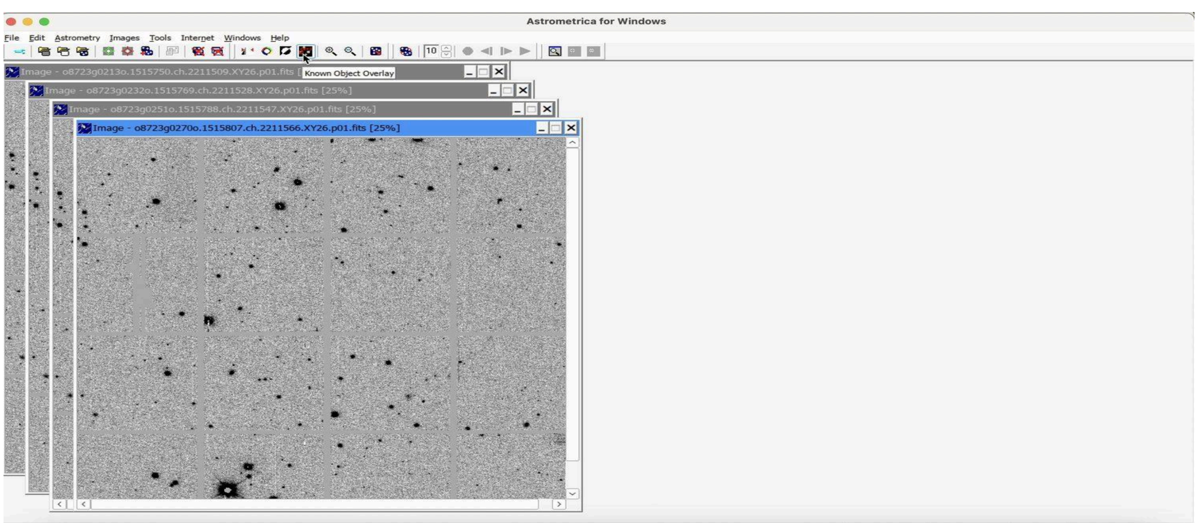

Figure 7a. *The Known Object Overlay control on the Astrometrica toolbar.*

Selection of this control cross-references the local MPCOrb.dat catalog against the timestamps and coordinates of the loaded frame.

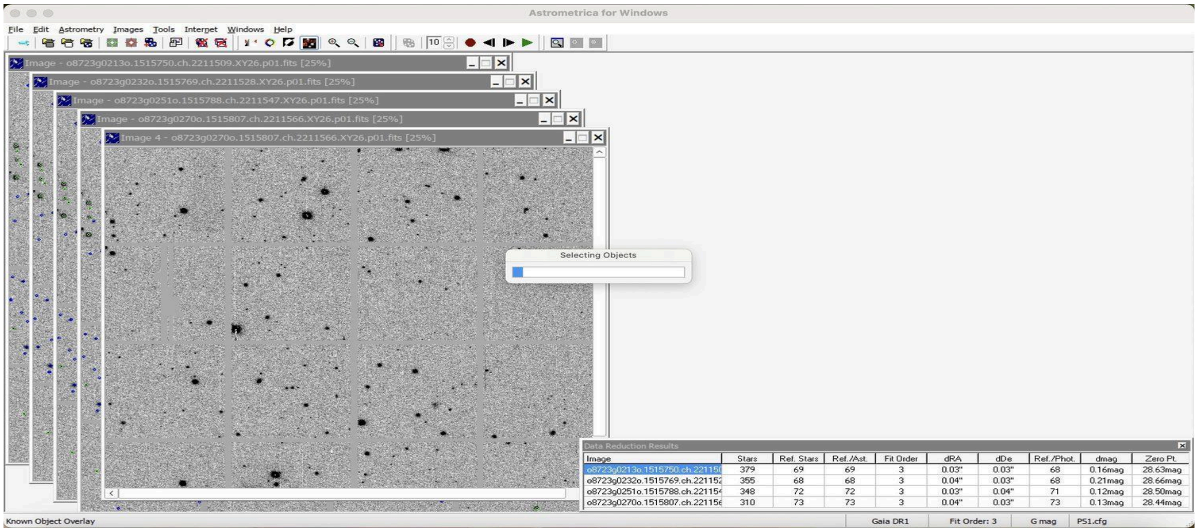

Figure 7b. *Known Object Overlay cross-referencing frame timestamps and coordinates against the local MPCOrb.dat catalog.*

On completion, previously catalogued objects are marked with a small red circle and labeled with their designation, consisting of either a permanent number (for example, 4643) or a provisional designation (for example, K19H08W).

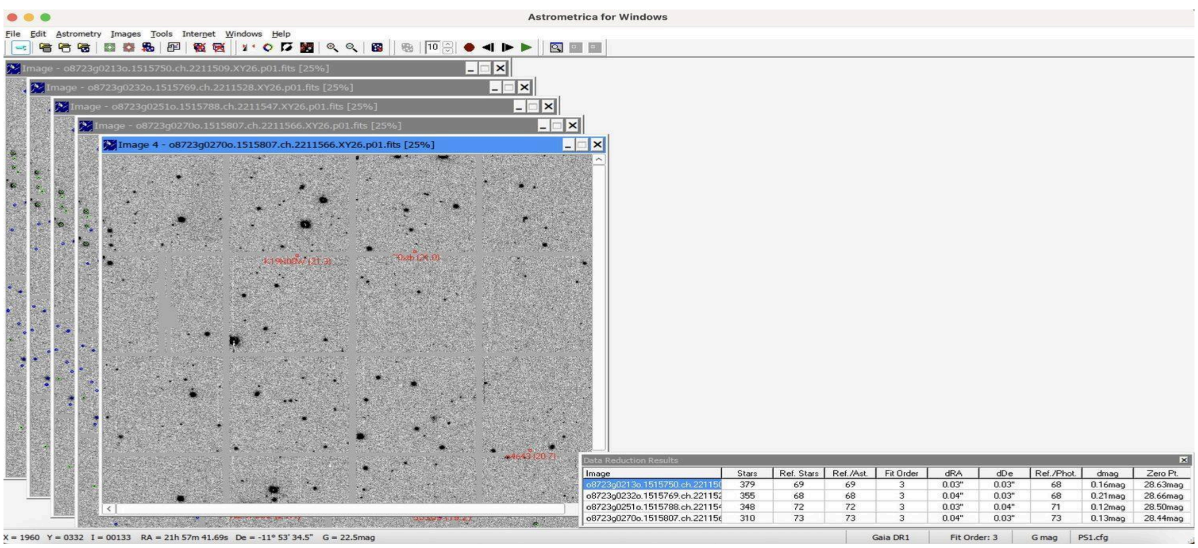

Figure 7c. *Known objects marked with red circles and their corresponding MPC designations following completion of the overlay.*

2.10 Manual Search Procedure

The manual search constitutes the primary observational component of the procedure. Unlike the preceding automated steps, it depends entirely on the reviewer's ability to distinguish a genuine moving object from noise, and it is the step most directly responsible for candidate discovery.

The Blink Current Images control on the toolbar (or the shortcut Ctrl+B) cycles through the four loaded frames in sequence, enabling detection of motion against the fixed background star field.

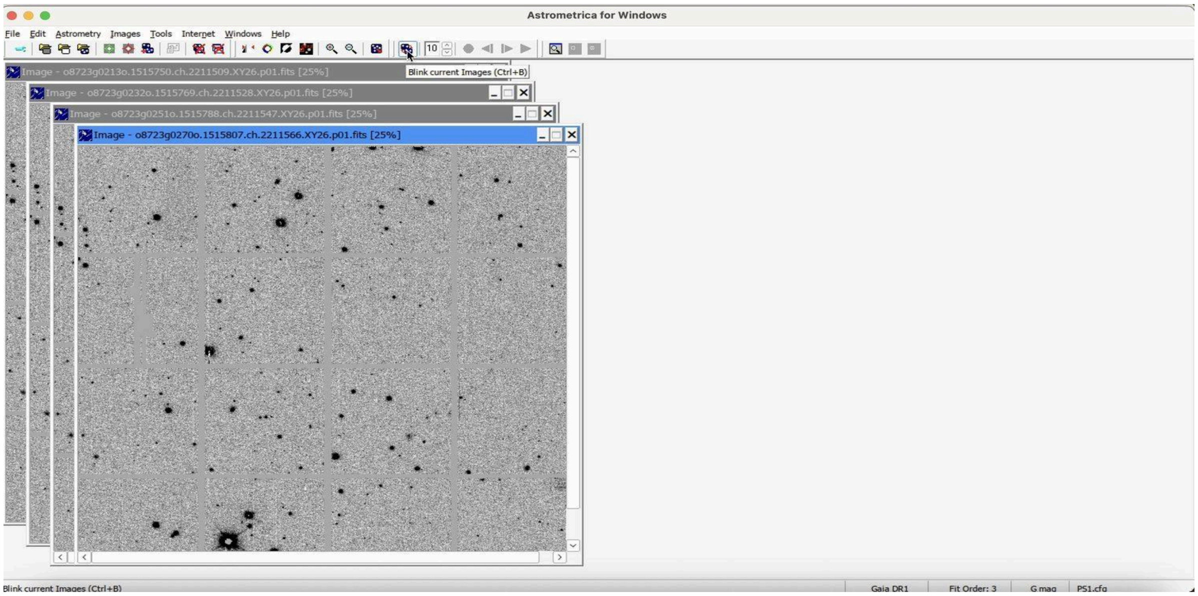

Figure 8. *Blinking of the four loaded frames in sequence, used to detect motion against the fixed background star field.*

2.10.1 Blink Configuration

- **Blink rate.** Adjustable from 1 (slow) to 9 (fast). A rate of 5 to 6 is a reasonable initial setting; a slower rate is recommended where distinguishing genuine motion from noise proves difficult, or within densely populated fields.
- **Zoom level and field coverage.** Systematic review of the frame, rather than examination as a single unit, is recommended. Dividing the image conceptually into a four-by-four grid of sixteen tiles and reviewing each tile in sequence, proceeding left to right and top to bottom, reduces the likelihood of an area being overlooked.
- **Invert Image.** Toggling between white on black and black on white displays can improve visibility of faint moving objects against background noise.

2.10.2 Candidate Identification Criteria

Three criteria are applied to determine whether a detected source constitutes a viable candidate.

- **Rectilinear motion.** Across four frames captured at intervals of several minutes, a genuine object exhibits motion along a straight line. Irregular or discontinuous motion indicates an artifact rather than a real object. Rectilinear motion is a consequence of the short observation window relative to the object's orbital period.

- **Constant angular velocity.** The displacement of the object should remain approximately consistent between successive frames. An abrupt change in displacement typically indicates a cosmic ray event or sensor artifact rather than an asteroid.
- **Point spread function consistent with a stellar source.** A genuine object should present as a round point source, consistent in appearance with the surrounding field stars, rather than as a streak, smear, or isolated hot pixel.

In addition to morphology and motion, the signal to noise ratio (SNR) reported upon selection of the object provides a quantitative indication of detection confidence. Values above approximately 3 to 3.5 are considered usable, and values above 5 indicate a strong candidate. Visibility of the object across at least three of the four frames is also a favorable indicator, as a source appearing in only one frame is more frequently attributable to noise than to a genuine detection.

3. Candidate Marking and Reporting

Where a detected source satisfies the criteria described in Section 2.10.2 and is not flagged by the Known Object Overlay, it constitutes a viable candidate for reporting. The following procedure is used to log a detection.

- The candidate is selected by double-clicking directly on the object within the blink view, opening the Object Verification panel.

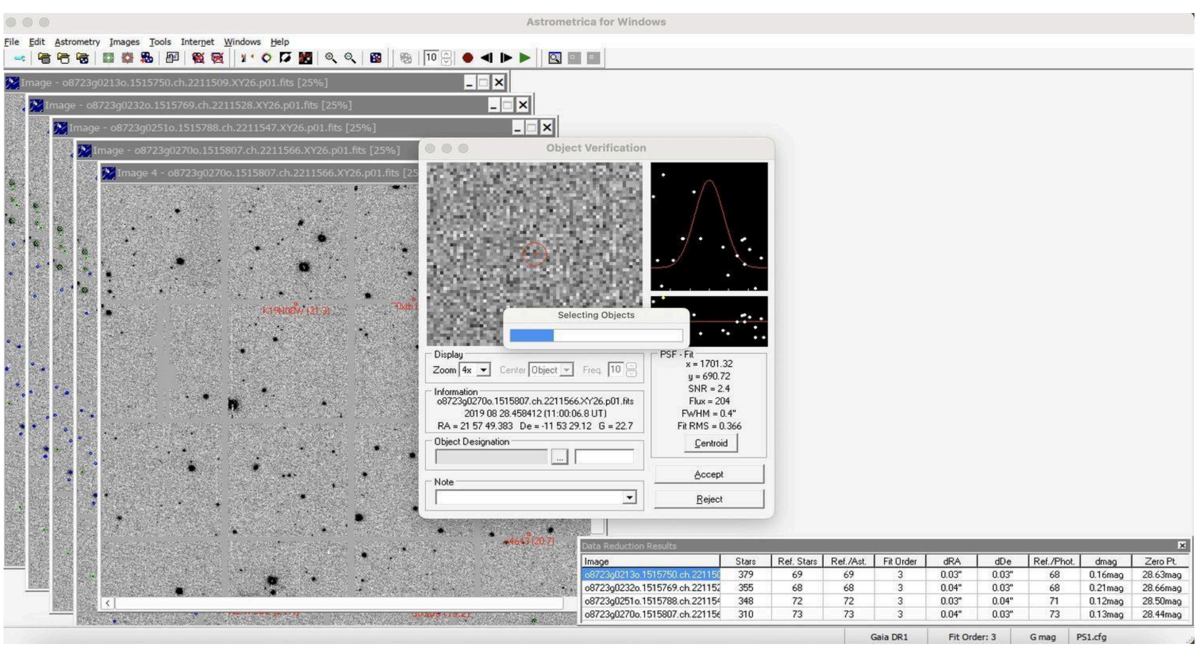

Figure 9a. *The Object Verification panel, opened by selecting a candidate within the blink view and displaying its point spread function fit and signal to noise ratio.*

- Astrometrica compares the selected coordinates against the local catalog and confirms whether the source is uncatalogued.

- The assigned object designation code is entered in the Object Designation field. The default placeholder text present in the software ('SPACE XXX', visible in the accompanying figures) is replaced with the code assigned by the campaign.

Note. *The designation code format consists of a three letter IASC Team Suffix, identifying the reporting team, followed by a four digit sequential number unique to the specific object within that image set. The same complete code is applied to a given object across all four frames; a second, unrelated moving object within the same set is assigned a separate sequential number.*

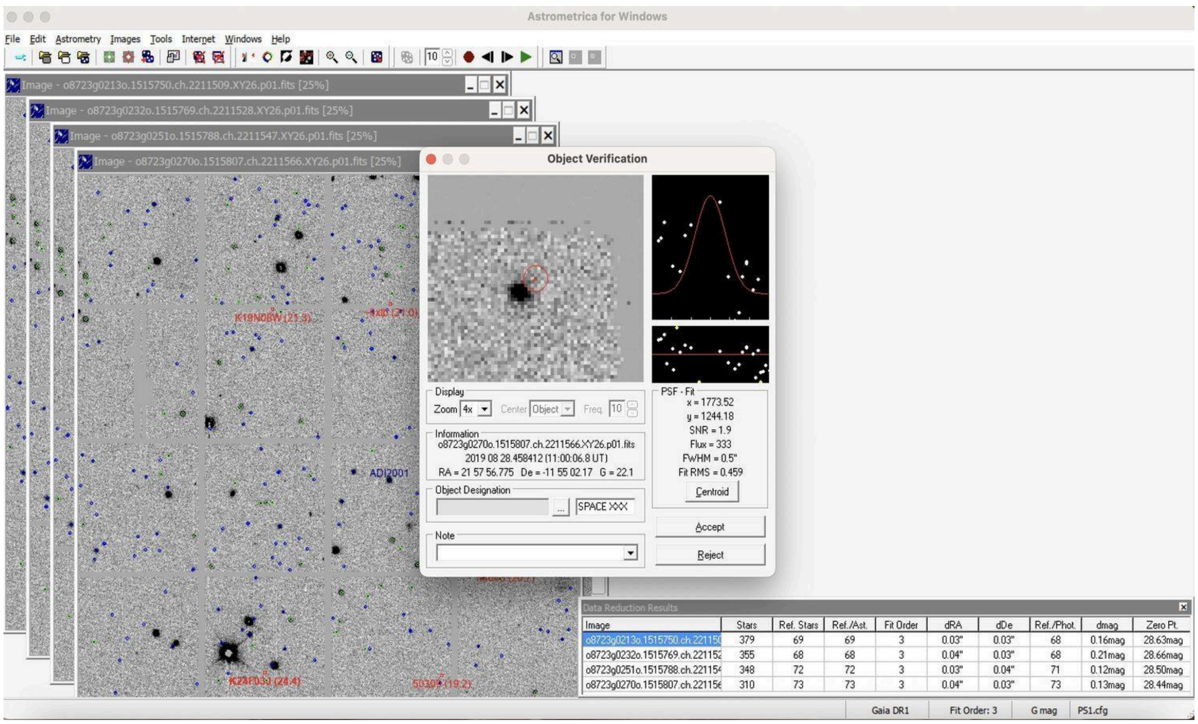

Figure 9b. *Entry of the object designation code prior to acceptance of the detection.*

- Additional notes required by the campaign are entered, followed by selection of Accept to log the detection for the frame.
- This process is repeated across all four frames using the identical code, as Astrometrica requires the object to be marked in all four frames to compute its trajectory.

3.1 Report Verification Prior to Submission

The generated report is reviewed under File > View MPC Report File prior to submission. Verification includes confirmation that the four timestamps are correctly ordered and that right ascension and declination values progress smoothly and consistently across frames, as an inconsistency in sequence typically indicates a marking error.

Three reporting scenarios exist, each requiring a distinct report format.

- **A previously uncatalogued object is identified.** The object is marked in all four frames using the assigned object code, as described above, and the generated report is submitted without modification.
- **No new object is identified, but a known object is visible.** Confirmation of a catalogued object's position remains useful data, as it verifies correct operation of the reduction and detection pipeline. The object is marked but accepted under its existing designation rather than an assigned code.

Note. *Where Astrometrica overlays a known object, the designation field is automatically populated using the catalog format, and this value should be verified for correctness prior to submission. Numbered asteroids require the standard five digit, zero padded format (for example, 04643, rather than 4643, notwithstanding that the on screen label is displayed unpadded), and provisional designations such as K19H08W require exact reproduction of capitalization and character sequence. An incorrectly formatted designation may cause the Minor Planet Center automated parser to reject the corresponding line or associate it with an incorrect minor planet, and a visual check of the report file prior to submission is therefore recommended.*

- **No moving object is identified.** An empty submission is not recommended, as it does not indicate that the frame is reviewed. The report line is instead replaced with a quoted note, for example “No moving objects”, to document that the frame is examined.

The completed report text is copied from the report file and submitted through the campaign submission portal.

3.2 Reset Between Image Sets

The File > Reset All Files command in Astrometrica is required between separate image set searches, i.e. when the current set of four images is replaced with a new set corresponding to a different sector of sky. It is not typically required between individual frames within the same set, since the software is designed to retain those four frames in memory for blink comparison and trajectory calculation. The reset is required between searches for the following reasons.

- **Cache clearing and data contamination.** Loading a new sky block without resetting can cause the internal object matrix maintained by Astrometrica to retain coordinate data from the previous sky sector. This produces spurious detections in which the software attempts to reconcile the new image coordinates against the previous star field map, causing the astrometric reduction to fail, typically reported as a “No solution” error.
- **Minor Planet Center report integrity.** The MPC report file functions as an appending text log. Without a reset, observations for the next target are appended to the same file as data from the previous target, producing a report that combines two unrelated targets and is likely to be rejected by the IASC submission portal.

- **Memory management.** Because Astrometrica executes through a Wine/Rosetta compatibility layer (via Sikarugir), the host system already performs substantial work to translate x86 instructions. Resetting clears the memory buffer and provides the execution environment with a clean state, reducing the likelihood of a crash during the subsequent astrometric reduction.

Reset is not required in the following case.

- **During a single sequence.** While processing of the current set of four images remains in progress, reset is not performed. The images and plate reduction data remain loaded throughout this process to allow continued blinking and verification of a candidate object.

4. Known Limitations

The following limitations apply to the procedure described in this report and should be considered when adapting it to other hardware, software versions, or campaign contexts.

- **Operating system compatibility risk.** Rosetta 2 and macOS Gatekeeper security requirements are subject to change across macOS releases. A configuration verified on a specific macOS version is not guaranteed to function identically following a future operating system update, and re-verification may be required after major macOS upgrades.
- **Reduced performance relative to native Windows.** Execution through the Wine/Rosetta compatibility layer introduces measurable overhead relative to native Windows execution, particularly during astrometric data reduction and moving object detection. Total processing time per image set is correspondingly longer than on native Windows hardware.
- **Dependency on third-party wrapper maintenance.** Sikarugir is a community-maintained project rather than a tool supported by Apple. Continued compatibility depends on ongoing maintenance by the Sikarugir project, and discontinuation or a breaking change could require adoption of a revised compatibility approach.
- **Single configuration verification.** This procedure is verified through manual execution of each step on the configuration specified above. Behavior on other Apple Silicon generations, other macOS versions, or other Sikarugir versions is not independently confirmed and may differ.
- **Scope limited to practice data.** Validation of the workflow described in this report uses IASC practice image sets rather than live campaign data, and does not constitute confirmation of behavior under live submission conditions, including MPC report portal validation.

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6. References

Astrometrica Official Documentation. (n.d.). *Astrometrica User Guide* (latest version). http://www.astrometrica.at/

Gaia Collaboration. (2016). The Gaia mission. *Astronomy & Astrophysics*, 595, A1. https://doi.org/10.1051/0004-6361/201629272

Greisen, E. W., & Calabretta, M. R. (2002). Representations of celestial coordinates in FITS. *Astronomy & Astrophysics*, 395(3), 1061–1075. https://doi.org/10.1051/0004-6361:20021326

International Astronomical Search Collaboration (IASC). (2026). Astrometrica software and campaign resources. https://iasc.cosmosearch.org/Home/Astrometrica

International Astronomical Union. (2006). *Definition of a Planet in the Solar System*. IAU Resolution B5. https://iauarchive.eso.org/static/resolutions/Resolution_GA26-5-6.pdf

Kaiser, N., et al. (2010). Pan-STARRS: A Large Synoptic Survey Telescope Array. *Proceedings of the SPIE*, 7733, 77330E. https://doi.org/10.1117/12.857304

Minor Planet Center. (2026). *MPC Orbit (MPCORB) Database*. https://www.minorplanetcenter.net/iau/MPCORB.html

Sikarugir Project. (2026). *Sikarugir: A user-friendly Wine wrapper for macOS*. GitHub repository. https://github.com/Sikarugir-App/Sikarugir